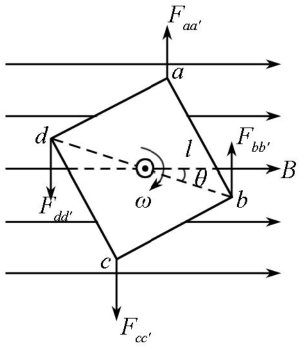
2. At any orientation, the angle of diagonal $db$ to the field is $\theta$. In $a'a$ 、 $cc'$ , and $b'b$ 、 $dd'$, the currents are

$$I_{a'a}=I_{ad}-I_{ba}=\frac{\sqrt{2}}{4}\frac{Bl^2\omega}{R}\sin\theta \qquad (10)$$

$$I_{cc'}=I_{bc}-I_{cd}=\frac{\sqrt{2}}{4}\frac{Bl^2\omega}{4}\sin\theta \qquad (11)$$

$$I_{b'b}=I_{ba}+I_{bc}=\frac{\sqrt{2}}{4}\frac{Bl^2\omega}{R}\cos\theta \qquad (12)$$

$$I_{dd'}=I_{ad}+I_{cd}=\frac{\sqrt{2}}{4}\frac{Bl^2\omega}{R}\cos\theta \qquad (13)$$



The torque needed for uniform rotation is the net magnetic torque on the circuit, so

$$F_{a'a}=F_{cc'}=BlI_{a'a}\text{，}\quad F_{b'b}=F_{dd'}=BlI_{b'b} \qquad (14)$$

$$M=2F_{a'a}\frac{\sqrt{2}}{2}l\sin\theta+2F_{b'b}\frac{\sqrt{2}}{2}l\cos\theta \qquad (15)$$

Substituting (11)~(14) into (15) gives

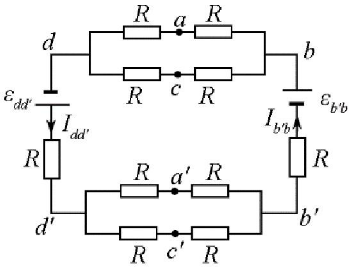
$$M=\frac{B^2l^4\omega}{2R}\text{。} \qquad (16)$$

Group scores: (10)~(13)、(4)、(5)、(2) : ; : ; : ; :

1 (2 total; (16)(15)(14)) : ; : ; : .

Method 2:

1. The induced emfs in $a'a$ and $cc'$ are



$$\varepsilon_{a'a}=\varepsilon_{cc'}=\frac{\sqrt{2}}{2}Bl^2\omega\sin\theta \qquad (1b)$$

The induced emfs in $b'b$ and $dd'$ are

$$\varepsilon_{b'b}=\varepsilon_{dd'}=\frac{\sqrt{2}}{2}Bl^2\omega\cos\theta \qquad (2b)$$

With $\varepsilon_{a'a}$, $\varepsilon_{cc'}$ shorted, only $\varepsilon_{b'b}$, $\varepsilon_{dd'}$ act. If $a'a$, $cc'$ are opened, the circuit is as shown. The currents in $b'b$ and $dd'$ are

$$I^{(1)}_{b'b}=I^{(1)}_{dd'}=\frac{\varepsilon_{b'b}+\varepsilon_{dd'}}{4R}=\frac{\sqrt{2}}{4}\frac{Bl^2\omega}{R}\cos\theta \qquad (3b)$$

The currents in $ba$ 、 $ad$ 、 $bc$ and $cd$ are

$$I^{(1)}_{bd}=I^{(1)}_{ad}=I^{(1)}_{cd}=\frac{1}{2}I^{(1)}_{b'b}=\frac{\sqrt{2}}{8}\frac{Bl^2\omega}{R}\cos\theta \qquad (4b)$$

By symmetry, the potential differences across $a'$ 、 $a$ and across $c$ 、 $c'$ are